

Data scientist focused on applied machine learning, NLP, and recommender systems, with experience building and evaluating models on multimillion-record datasets and developing scalable data processing pipelines.

Education

M.S. in Data Science <i>University of California, San Diego</i> University GPA: 3.81	2026 La Jolla, CA
B.S. in Management Information Systems <i>California State University, Sacramento</i> University GPA: 3.80 (Magna Cum Laude)	2024 Sacramento, CA
A.S. in Accounting A.S. in Business Administration	

Projects

Aspect-Based Sentiment Analysis of Google Reviews (2026) Compared classical NLP, transformer, and LLM approaches for aspect-level sentiment extraction across 22M+ real-world Google reviews.	(Capstone Project, Paper , GitHub)
- Engineered 14 multi-label targets using weak supervision with star ratings as proxy labels across 70 million California reviews (25GB+) - Benchmarked TF-IDF + Logistic Regression, fine-tuned RoBERTa, and GPT-4.1-mini zero/few-shot prompting for aspect-level sentiment classification - GPT-4.1-mini zero-shot prompting produced strongest ground-truth performance (Weighted F1: 0.757), RoBERTa performed best on weakly labeled data (Weighted F1: 0.930), highlighting label-noise limitations in weak supervision	
Amazon Sports Products Recommender System (2025) Hybrid recommender system designed to surface relevant products from a sparse 4M+ Amazon review dataset.	(Academic Project, GitHub)
- Mitigated 99.998% sparsity by combining collaborative filtering (SVD) with content-based features (Ridge regression) to improve cold-start performance - Benchmarked five recommendation strategies and optimized hybrid weighting (60% SVD, 40% Ridge) to balance personalization and coverage - Achieved strong ranking quality (nDCG@10 0.9981) with 31.5% RMSE reduction over baseline	
NYC Yellow Taxi Fare Prediction (2025) Fare prediction model built from 150M+ NYC taxi records, earned 3rd place in class competition.	(Group Project, My Contributions GitHub)
- Engineered time/efficiency features from 20GB+ trip data using Apache Spark and Dask - Trained regressor achieving MAE \$1.91 and R² 0.94, improving RMSE by 49% over baseline - Scaled training with distributed computing (15 workers, 60 cores) and validated performance using residual analysis	

Skills

Programming: Python, SQL, JavaScript (D3.js)
Machine Learning: Scikit-learn, PyTorch, Hugging Face Transformers, XGBoost, LightGBM, Recommendation Systems, NLP, Hyperparameter Tuning
Data Processing: Pandas, NumPy, Apache Spark, ETL Pipelines, Large-scale Data Processing
Tools: Git, Tableau

Experience

DBI Beverage <i>Operations Team Lead</i>	2014 - 2020 Stockton, CA
- Managed 7-person team distributing 50K+ cases/day for Coors, Modelo, and Corona brands - Spearheaded new Warehouse Management Software (VIP EasyOps) implementation, modernized workflows - Improved inventory accuracy 12% through better labeling, FIFO procedures, and discrepancy investigation	